SSHVisual: Visual Tool to Analyze SSH Data

Shreeman Gautam
Kahlert School of Computing
University of Utah
Salt Lake City, UT, USA
email address or ORCID

Robert Ricci
Kahlert School of Computing
University of Utah
Salt Lake City, UT, USA
email address or ORCID

***Abstract*—Network operators use visualizations to detect attacks on their systems, interacting with data to uncover patterns that might stay hidden. Operators use these findings to block attackers and to expand the knowledge base of known threats. Detecting anomalies and identifying relationships iteratively in data is, at best, tedious and, at worst, infeasible without data visualizations. Prior work in network visualization analyzed the IP address and ASN dimensions of SSH logins with little focus on the username aspect of the attackers and detected anomalous network events when baseline events were available. As a result, little attention has been paid to whether attackers that differ in timing, source address, and username belong to the same coordinated campaign. We present SSHVisual, an iterative visualization system evaluated on login attacks collected from our production SSH systems. SSHVisual analyzes the IP, ASN, and username dimensions of SSH attacks, to support campaign detection. It complements automated detection by allowing operators to find anomalous events without a normal baseline. It uses a similarity search across dimensions that groups attackers by behavior and explains each match. It supports iterative and collaborative sharing of findings between operators. We evaluated SSHVisual on 206.6 million attacker login attacks collected over 69 days. We found that it helps operators surface hidden patterns and find attackers to block. In one case study, some IPs tried a similar set of usernames, including uncommon ones, which pointed to a single obscure campaign that volume charts could present as unrelated bursts. In another, similarity search surfaced low-volume IPs working through various username lists, none of which ranked in the top ten by volume. Without similarity search, these IPs would likely have been hard to find and block. These results suggest that username-centric, operator-driven exploration can be valuable because it can reveal covertness and coordination that previous analyses can miss.**

***Index Terms*—SSH, brute-force attacks, attack campaigns, network security visualization, visual analytics, similarity search**

I. Introduction

Computer systems employ SSH [1] to allow users to securely connect to them in order to execute commands and do remote administration. Many users do a username, password combination (password based authentication) in order to gain entry to the SSH system. Attackers, who are adversarial, can use brute force attack mechanisms [2] to get into SSH systems. Sometimes, they guess the username, password combinations and other times, they take advantage of poorly secured systems to build botnets [2] to carry out further attacks [3], [4] or steal resources from the systems, like cryptomining [5], [6].

Given the prevalence of SSH attacks in the IPv4 space [2], we designed SSHVisual and evaluated it on login attacks from our production SSH systems. SSHVisual serves two kinds of network operators: Network administrators, who handle daily SSH traffic, need to identify attackers quickly and block them and Security analysts, who study traffic over longer periods, look for patterns that are not apparent and share what they find. Consider a single period: an administrator can isolate the highest-volume attacker and block it, while an analyst examining the same window can iteratively surface a set of low-frequency attackers that attempt an identical sequence of usernames, and group them as a coordinated campaign.

Surveys of network visualization [7]–[11] establish the maturity of the field and the gaps we target. Our two contributions follow from treating the username, alongside the source IP and ASN, as part of an attacker's identity, and each one serves one of the two operators described above. First, for security analysts, we show that username-based similarity identifies coordinated campaigns that volume-based views split apart. IPs that try the same list of usernames can be recognized as a single campaign, even when they come from different countries and ASNs and are never active on the same day. Second, for network administrators, we show that behavioral similarity search finds attackers that would otherwise stay hidden. Starting from a single IP the administrator already intends to block, it surfaces low-volume IPs that act alike but never reach the top-ten volume rankings, and it explains each match, so the administrator knows which other IPs to block and why. Since network traffic may not offer a stable, attack-free baseline of normal behaviour [12], detectors that rely on such a baseline may miss activity that doesn't deviate from it in the expected way. Operators, by contrast, may recognize coordination and similarity across sessions visually, without specifying normality in advance. The above contributions build on this ability and are meant to complement baseline-driven detection. As operators explore, they can label and share findings so colleagues do not investigate the same patterns twice, and the findings can in turn inform automated detection.

The rest of the paper is as follows: Section II is Related Work, Section III is Tasks and Requirements, Section IV is Design, Section V is Evaluation and Section VI is Conclusion.

II. Related Work

A long line of visualization tools [13]–[22] renders network traffic and security events navigable for operators, the majority being network-flow and traffic oriented [13], [14], [16],

[18], [19], [21], [22], with some also integrating intrusion-detection/prevention and host logs [15], [17] or multidimensional cyber-defense data [20]. Additionally, [23] visualizes IDS alerts and [24], [25] show heterogeneous security logs. [26] extract multi-resolution discrete-wavelet features from traffic, project them with principal component analysis, and apply hierarchical clustering across several intrusion-detection benchmarks, separating brute-force attacks into distinct clusters. Related work on coordinated multiple views [27]–[29] and insight generation [30] informs our design.

A. Username Campaigns with Time

To our knowledge, honeypot literature has rarely treated the username identity as an actor for campaign detection purposes. Typically, it is a dimension to count rather than to track, whereas in our analysis it serves, alongside the IP and ASN, as an anchoring dimension. Several papers [31]–[37] collect or chart honeypot traffic, with several of them capturing SSH credentials (usernames and passwords). Where credentials are charted, like the SCADA SSH honeypot [34], kippo-graph [35], and the container deployment [36], usernames and passwords appear as static top-N tallies (kippo-graph's top-10 username and password bars being an example), and not placed on a time axis. The WAN honeypot framework [38] is the closest prior system : it computes an IP-to-IP dictionary-overlap (containment) score to probe whether six same-/24 source IPs share a wordlist, which is a manual, one-off prototype of credential-based coordination analysis, performed on the IP and without a temporal dimension.

B. Username-Based Similarity

Computed similarity between entities is well established in literature [39]–[42]. We do not claim similarity computation itself as novel; the contribution lies in the entity it operates on. Prior systems cluster internal hosts [39], behavioral profiles [41], IP/port members [42], or internal enterprise users grouped by shared resources [40]. None treat the username as an attacker entity. We use the username as a signature of an IP, grouping IPs by the credentials they try, with uncommon usernames weighted as stronger evidence than ones nearly every attacker tries. Honeypot/honeynet analyses have related attack activity through time by grouping attacks whose time series rise and fall together [43], [44]. Attackers that share a credential list, however, need not be active together, so we compute similarity independently of when each attacker is active and let the operator see their time series in a linechart.

C. Limits of Automated Detection

Automated anomaly detection assumes a normal baseline, and SSH traffic generally does not have one. Even where a baseline exists, such detection is fragile. [45] find that machine-learning anomaly detection is rarely deployed, because normal traffic varies widely, errors are costly, and sound evaluation is hard. [46] show that when intrusions are rare, the false-alarm rate limits any intrusion detector. Our data is the opposite case, since nearly every login attempt is an attack. [12] question whether attacks are rare and whether attack-free training data exists at all, and note that malicious traffic can look normal. [39] and [47] learn a model of each actor's expected behavior with [47] assuming that an actor's activity repeats weekly. However, constantly churning, near-uniformly hostile traffic like we see cannot offer a per-actor normal. Situ also concedes that operators "do not want another automated tool with algorithms they do not trust" and keeps a human in the loop. Lacking a per-actor baseline, we make campaign identification operator-driven from the start.

D. Username as Actor Identity

We analyze attackers at the level of a campaign, meaning a group of attackers acting together, rather than one attacker at a time. That raises a practical problem: a campaign rarely shows up as a single source IP. IPs may not be tied to one machine. One IP can be assigned to different hosts [48], hosts can cycle through new IPs [49], and counting by address can overstate the number of real machines sometimes by an order of a magnitude [50]. A view organized by address is therefore likely to scatter one campaign across dozens of entries that can look unrelated. The username can pull those pieces back together. Attackers in the same campaign likely draw from a shared credential list, so the same login names may recur across changing IPs. Grouping by username can reassemble the campaign that an address-by-address view could split apart.

E. Operator Labeling and Collaboration

The findings an operator produces can be captured and shared. In two works [19], [51], interactive labeling was used to fold the operator's judgments back into the system. Collaborative analysis was supported through shared artifacts with [17] and [20] being examples. These systems establish that labeling and collaboration are valuable, which is why our labeling-and-sharing mechanism is the step that turns an identified set of usernames into a durable, shared campaign.

III. Tasks and Requirements

The design of SSHVisual draws on our experience as network operators and on gaps we identified in prior work on SSH attack analysis and security visualization. From these we derive three tasks and five requirements.

A. Tasks

T1: Identify SSH campaigns. Security analysts need to find attackers that behave alike across usernames, IPs, ASNs, and countries, and record groups so they can be referenced.

T2: Detect anomalies that warrant action. Network administrators need to see unusual activity, like a sudden spike or a source dominating traffic, and decide whether to block it.

T3: Share findings across operators. Analysts and administrators need to see what their colleagues have already investigated, so that the same patterns are not identified twice.

B. Requirements

R1: Attacker similarity profiling. SSHVisual should be able to cluster similar attackers along the username, IP and ASN dimensions. R1 mostly serves T1 as it allows security analysts to build a repository of similar attackers.

R2: Sorting and ranking. SSH data can be heavy-tailed: one username can account for a quarter of attacks (remaining attacks spread across other usernames) [52] and persistent IPs can account for most attacks [53]. SSHVisual should therefore list every entity in each dimension, and let the operator sort and rank entities by any combination of attributes. R2 supports T1 and T2: network administrators can surface anomalous entities that may not reach the top of a volume ranking, and security analysts can find similar entities.

R3: Shared tagging. Operators should be able to tag attackers, record what a tag means, and why it was applied to a particular attacker, and have those tags remain visible to every operator. Each tag application should record who applied it and when, so later operators can judge whether a finding is still current. R3 primarily supports T3, and supports T1 because tags accumulate into a repository of related attackers.

R4: Complementary tabular and temporal views. No single visualization can serve every need. A table can support precise lookup and ranking but can discard temporal structure; a linechart can preserve temporal structure, showing spikes and campaigns at a glance, but can convey values less precisely. SSHVisual should therefore provide both views and let operators move between them. A network administrator may find and act on an anomaly using the table alone, whereas a security analyst may use the table to assemble related attackers and examine their behavior over time in the linecharts. R4 supports T1 and T2.

R5: Interactive exploration of the linecharts. The linecharts follow Shneiderman's Visual Information-Seeking Mantra: overview first, zoom and filter, then details-on-demand [54]. They open on the full time range, let operators zoom into a time window and filter entities, and reveal an entity's details on request. R5 supports T1 and T2: zooming can isolate the time window of a suspected campaign, and filtering can remove dominant entities that would mask smaller ones. Line charts can become harder to read accurately as more series are added [55]. Each linechart should therefore show few entities and we decided to cap it at 10.

IV. Design

SSHVisual combines two visual components, a discovery table and linecharts, with two non-visual components, similarity search and tagging. Section IV-A and Section IV-B describe them respectively. All figures show SSH login attacks between Nov 1 2022 and Jan 8 2023, a period of 69 days.

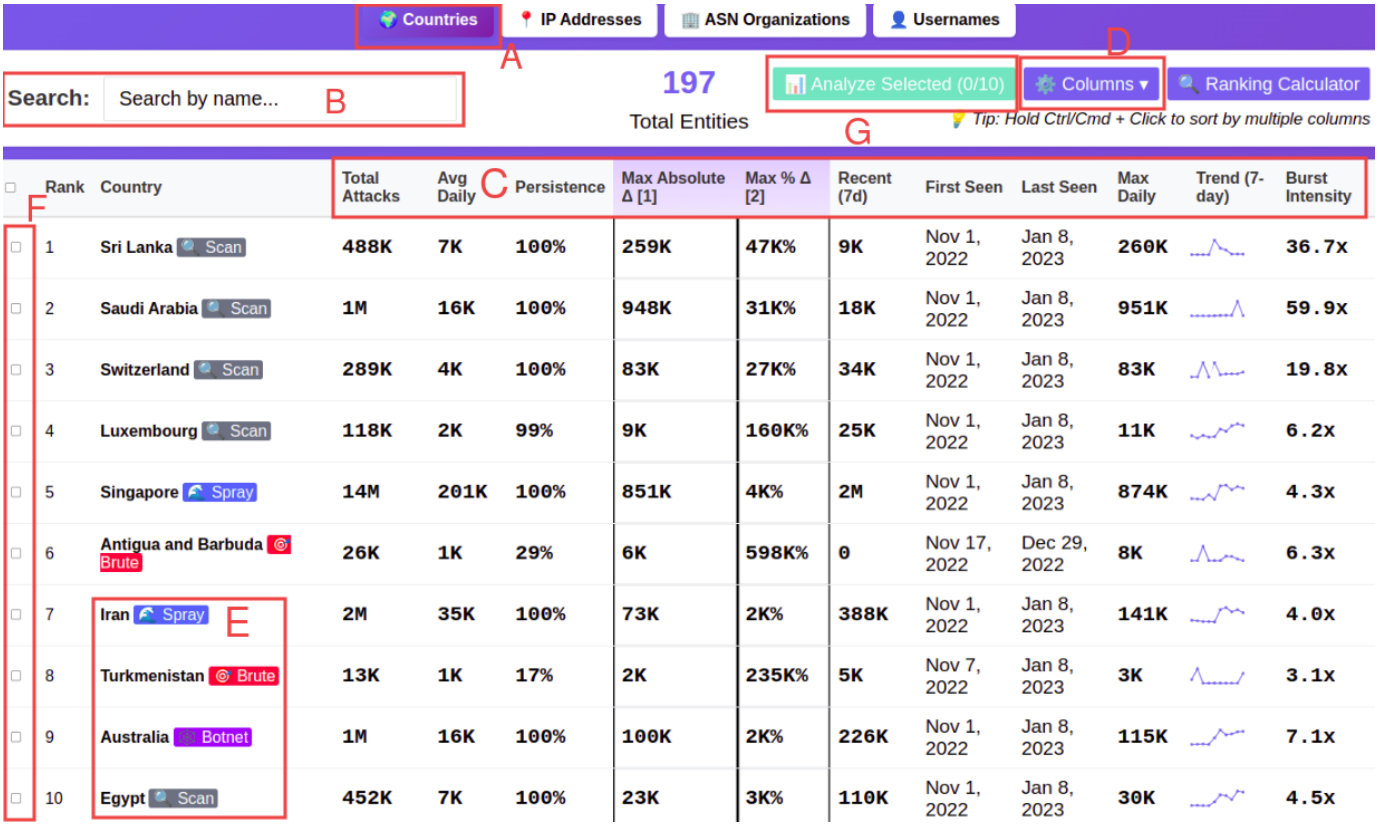


Fig. 1. Discovery table, showing the countries tab (A). Operators can filter countries(B). A row shows a country and operators can sort by any mix of columns (C), with more columns present in (D). Based on attack patterns, a badge is given (E). Operators can select up to 10 rows (F and G) to plot in linechart.

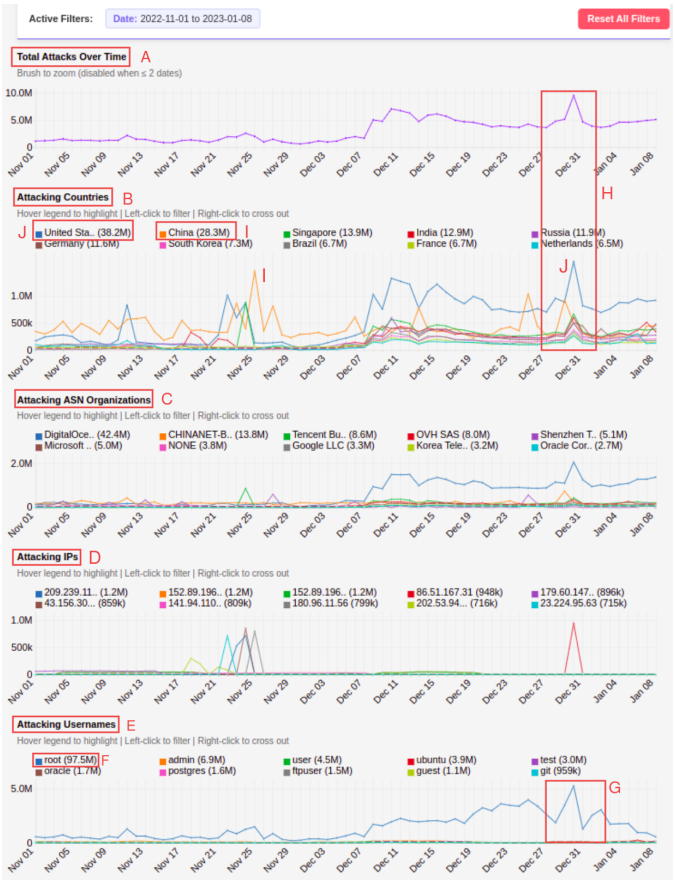


Fig. 2. The top chart (A) shows total attacks over time, and the four charts below it show the 10 most active countries (B), ASNs (C), IPs (D), and usernames (E). A legend entry (F) identifies each series, and a brushed region (G) selects a time window. A spike appears around Dec 30 (H), and two lines (I, J) follow individual countries' attacks over time.

A. Visual Design

The interface provides two views of the data. The discovery table (Fig. 1) is time independent: each row summarizes one entity over time. The linecharts (Fig. 2) can be time-dependent. Both views cover the full data period by default.

1) Discovery Table: The discovery table has four tabs, one per dimension: Countries, IP addresses, ASN organizations, and Usernames. In Fig. 1, the Countries tab is active (label A). A search box filters rows (label B), and the counter above the table reports the number of entities (197 countries). Some columns (label C) are shared across all tabs: total attacks, average daily attacks, persistence (the share of days on which the entity was active), the largest absolute and percentage daily change, attacks in the most recent seven days, first- and last-seen dates, maximum daily count, trend (line chart showing attacks at seven day intervals) and burst intensity (ratio of max daily attacks to average daily attacks where a higher ratio likely indicates an entity that attacks sporadically but with higher frequencies). The full list and figures of columns (labels C and D) are found in Appendix A.

Exclusive to the countries tab, each country also carries a badge summarizing its dominant behavior, namely Scan, Spray, Botnet or Brute (label E). Brute marks entities whose traffic comes from few IP addresses making many attacks using few usernames. Spray marks traffic from many IP addresses trying many usernames, and Botnet marks traffic spread across many IP addresses and many ASNs. Scan marks a general attack pattern that fits none of the other three.

Unlike the linecharts, which show only the 10 most active entities per dimension, the table lists every entity. Operators can select up to 10 rows with the checkboxes (label F) and use Analyze Selected (label G) to plot those entities in the linecharts, and they can return to the linecharts at any time. This link lets operators move from a precise, ranked lookup to the temporal behavior of the same entities, fulfilling R4.

2) Linecharts: The linechart view (Fig. 2) stacks five charts on a shared time axis. The first, Total Attacks, shows the aggregate volume of attacks (label A); the remaining four show the 10 most active countries, ASNs, IPs, and usernames (labels B-E). Each legend entry (label F) lists the entity's total for the selected period. Interaction with the linecharts follows [54] (fulfilling R5), and the charts are coordinated so that an action in any one of them updates all of them.

Overview first. The charts cover all 69 days of data, with each chart showing the 10 most active entities in its dimension.

Zoom. Every chart supports brushing (label G). When the operator brushes a time range on any chart, the range is applied to all five charts and recorded in the Active Filters bar, so the temporal context stays the same whichever dimension the operator is examining. Brushing is disabled once the range is two days or fewer, since a line needs more datapoints.

Filter. Right-clicking a legend entry crosses the entity out, removing it from its chart and rescaling the y-axis so that smaller series are no longer flattened against the axis by a dominant one. Left-clicking a legend entry selects that entity and filters every chart to the attacks associated with it. Because each attack links a username, an IP, an ASN, and a country, the selection propagates along these relationships:

- Selecting a country restricts the ASN, IP, and username charts to the entities observed in attacks from that country
- Selecting an ASN shows the country it originates from, along with its IPs and the usernames they try
- Selecting an IP shows its country and ASN, along with the usernames it tries
- Selecting a username shows the countries, ASNs, and IPs from which it was tried

Details-on-demand. Hovering over a data point shows a tooltip with that entity's attack count for the day, and hovering over a legend entry highlights the entity's line, and the legend reports each entity's total for the selected period.

Linked filtering can turn linecharts from four separate rankings into a means of tracing an entity across dimensions. An operator investigating the spike on Dec 30 (label H) can select root in the username chart; the IP, ASN, and country charts then show which addresses, networks, and countries tried root, and brushing around Dec 30 narrows the view to the spike. Conversely, selecting an ASN like DigitalOcean reveals

	Rank	Username	Total Attacks	Avg Daily	Persistence	Max Absolute Δ [1]	Max % Δ [2]	Recent (7d)	First Seen	Last Seen	Max Daily
☐	1	karaf	77.6K	5.2K	21.70% (15d)	32.4K	360.52K%	120	Nov 2, 2022	Jan 8, 2023	44.8K
☐	2	ossuser	82.6K	2.6K	46.40% (32d)	30.4K	276.15K%	690	Nov 3, 2022	Jan 8, 2023	46.0K
☐	3	snow	248.2K	5.2K	69.60% (48d)	22.0K	168.87K%	41.8K	Nov 1, 2022	Jan 8, 2023	35.2K
☐	4	nmrsu	89.4K	1.8K	72.50% (50d)	35.0K	145.91K%	2.9K	Nov 5, 2022	Jan 8, 2023	43.8K
☐	5	lenovo	25.3K	476.51	76.80% (53d)	8.5K	283.10K%	5.7K	Nov 1, 2022	Jan 8, 2023	8.5K
☐	6	fb	9.7K	225.67	62.30% (43d)	4.7K	470.10K%	403	Nov 5, 2022	Jan 8, 2023	4.7K
☐	7	inspur	40.4K	684.71	85.50% (59d)	5.3K	263.05K%	11.2K	Nov 1, 2022	Jan 8, 2023	5.3K
☐	8	radware	12.3K	686.11	26.10% (18d)	4.8K	241.25K%	1.9K	Nov 1, 2022	Jan 8, 2023	4.8K

Fig. 3. Usernames tab sorted by Max Absolute Δ (A) and Max % Δ (B), surfacing bursty usernames (C).

which usernames its hosts try, indicating whether a hosting provider's traffic is one campaign or many. Brushing on every chart, linked filtering, and on-demand details fulfill R5.

The linecharts answer questions that the discovery table cannot, such as "Do the countries in Fig. 1 share the same peaks and valleys?" or "Does one country begin attacking earlier than another?" In Fig. 2, the aggregate chart shows a spike on Dec 30 (label H), and most top-ten countries peak on the same day, indicating a coordinated, multi-country event rather than a single source. The charts also reveal timing differences. China's largest peak falls in late November (label I), weeks before the US' activity rises (label J), and the IP chart shows individual addresses that each spike for a day or two then fall silent. Neither pattern can be recovered from the table's per-entity totals. Together, Figs. 1 and 2 show how the two views complement each other, fulfilling R4.

3) Sorting and Ranking: Fig. 3 shows the usernames tab sorted on two columns (labels A and B). Operators sort by holding Ctrl/Cmd while clicking headers, and the bracketed indices in the headers show when the column was clicked.

Sorting matters because the data are heavy-tailed. The username linechart in Fig. 2 is dominated by root, which accounts for tens of millions of attacks and flattens every other line. Ranking by day-over-day change instead surfaces usernames (label C in Fig. 3), absent from Fig. 2, several of which are default accounts for specific software or hardware products (karaf, for example, is the default account of Apache Karaf). Their totals are far below the top ten, but their activity is concentrated: karaf was used to attack 78 thousand times, 45 thousand of them on a single day, and was active on only 15 of the 69 days. Bursts of product-specific credentials like these suggest campaigns targeting particular devices, yet they can be invisible in a volume-ranked chart. Fig. 3 thus shows how sorting and ranking over the full entity list fulfills R2.

B. Grouping and Tagging

The visual components help operators find individual attackers, while the grouping and tagging components help them connect those attackers to others and record what they find. Similarity search groups attackers by behavior, in the username dimension (Section IV-B1) and the IP dimension (Section IV-B2), and shared tagging (Section IV-B3) preserves the resulting groups so that other operators can build on them.

1) Username Similarity: Similarity search lets operators start from one attacker and retrieve the attackers that behave most like it. For usernames, the unit of analysis is the population of sources that tries a username. For a username, we compute the breadth of that population (the number of distinct IPs, ASNs, and countries that try it), its volume, traffic concentration on its most active source, among other features. We do not reuse the 17-dimensional profile we apply to IPs (Section IV-B2), because a diagnostic showed some features to be uninformative for usernames. Daily stability is an example: most IPs are active many days, so their stability is well defined, but half of the usernames appear only one day, where stability is meaningless. We cluster username profiles with k-means (k=9). Each cluster is labeled automatically from its centroid. To answer a query, the system retrieves the query's cluster and ranks the other members by Euclidean distance to the query, reported as a similarity score.

In Fig. 4, an operator searches for james (label A). The system places it in Cluster 3, labeled High-Volume Persistent Botnet-Scale, which contains 2,624 usernames; a cluster member receives about 31 thousand attacks from roughly four thousand IPs and is active on 72% of days (label B). The results table lists the 20 most similar usernames (five shown here), each with its total attacks, top country, top three source IPs, the top three ASNs, and a set of "Why similar?" labels (label D), such as a shared lead source country, comparable volume, or sourcing volume, that explain the match.

Above the results, a shared infrastructure panel (label C) lists IPs that appear as the top sources of more than one username, also shown in the table. In Fig. 4, 37 IPs are shared. Each panel line names an IP and counts how many of the 100 usernames it is a top source for, followed by examples of those usernames. Two of the most shared addresses, 152.89.196.123 and 152.89.196.220, lie in the same /24 network and are top sources for 34 and 33 usernames respectively, including manager, daniel, and oscar. The panel applies the same analysis to ASNs: 13 are shared, led by DigitalOcean (99 of 100 usernames), Tencent (90), and OVH (40). This panel can turn behavioral resemblance into infrastructural evidence: usernames tried by the same hosts are likely drawn from the same credential list and belong to the same campaign.

2) IP Address Similarity: We characterize each IP address by a 17-dimensional profile drawn entirely from its activity during the data period. The features fall into four groups:

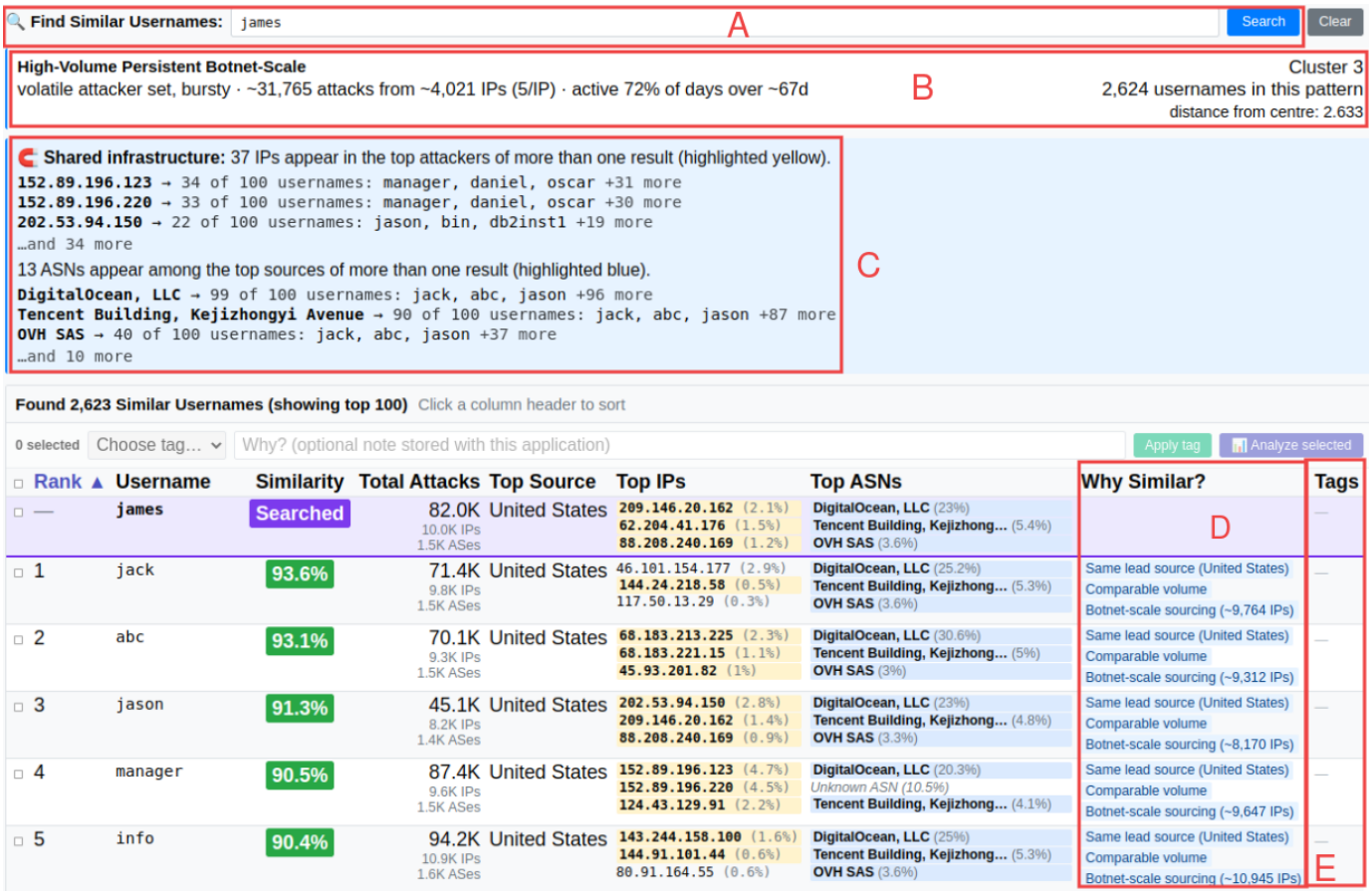


Fig. 4. Similarity search for james (A). Cluster summary (B), Shared infrastructure (C), and Why similar? (D) explain why the results were grouped.

volume (log-scaled totals, daily means, and peaks); temporal shape (the fraction of days on which the address was active, its burst intensity, its largest daily swings, a fitted trend slope, its activity span, and the share of its traffic in the final week); targeting behavior (unique usernames tried, how stable that set is from one day to the next, its rotation rate, and the share of attacks taken by its most-tried username); and two flags indicating whether the address belongs to a cloud provider and whether it originates from a major source country. We cluster these profiles with k-means, label each cluster from its centroid, and answer queries in the same way as for usernames.

In Fig. 5, an operator searches for 61.177.173.46 (label A), which falls in Cluster 5, labeled Low-Volume Persistent Exploratory, containing 4,405 addresses. The most similar results are shown in label B, with similarity scores between 80.4% and 91.2%, all belonging to CHINANET-BACKBONE. Behavioral features alone can recover infrastructure relationships, suggesting that these hosts may be operated together.

Figs. 4 and 5 show how SSHVisual lets operators assemble sets of similar attackers along the username and IP dimensions and see why those attackers were grouped, fulfilling R1.

3) Shared Tagging: Figs. 6–8 show how an operator records a finding so that other operators can see work that has already been done. Tags are organized in a tag repository for

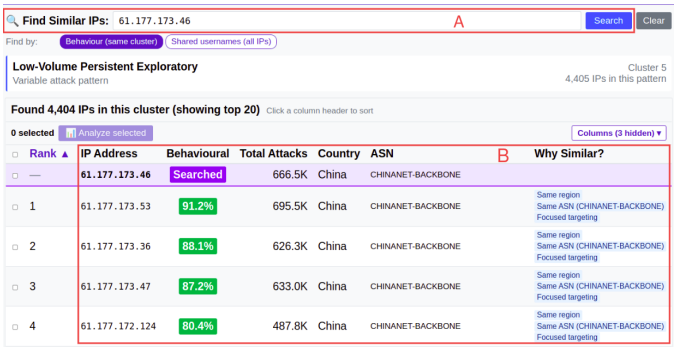


Fig. 5. Similarity search (A) for IP 61.177.173.46. Nearest neighbors are adjacent addresses on the same network (B).

each dimension. The repository displays the current operator's name, and every tag and tag application records its author.

In Fig. 6 the operator creates a tag, "similarity to james" (label A), for the username cluster found in Fig. 4. The repository lists the tag with its description (label B) and its author (label C). Tags are applied from the similarity results. In Fig. 7 the operator selects seven usernames (label A), chooses the new tag, and adds a note explaining this application: "recurring ASNs" (label B). In Fig. 8 the repository reports

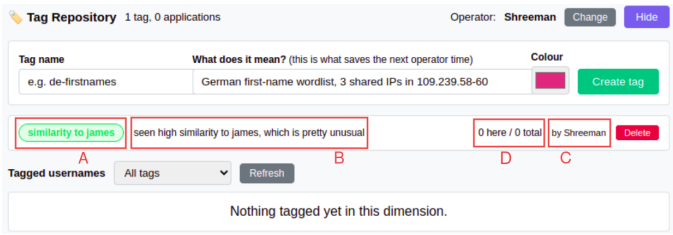


Fig. 6. A newly created tag (A), listed with its description (B), author (C), and application counts (D).

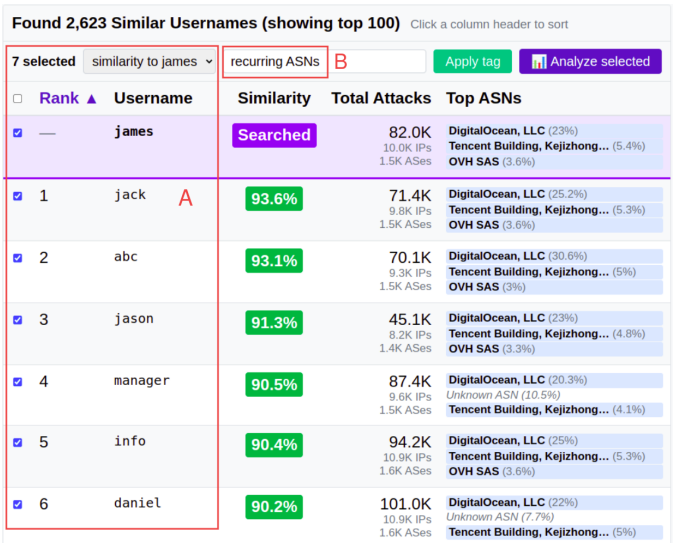


Fig. 7. Applying the tag to seven usernames from the similarity results (A), with a note explaining the application (B).

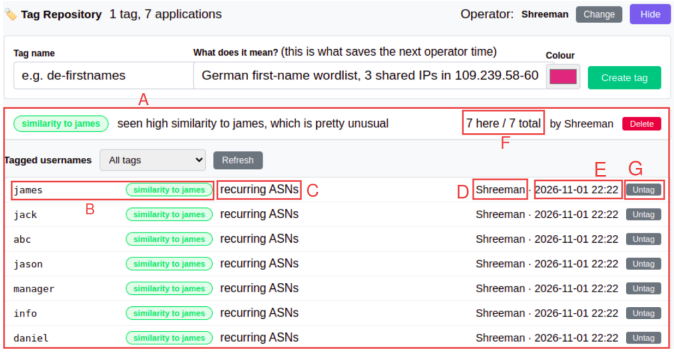


Fig. 8. Repository after tagging (A), listing tagged username (B) with note (C), author (D), timestamp (E), application counts (F), and untag option (G).

seven applications (labels A and F). Filtering the list by tag shows each tagged username with its note (labels B and C), the operator who applied it (label D), and a timestamp (label E); any application can be removed with Untag (label G). Tagged usernames also appear in the Tags column of future similarity searches (label E in Fig. 5), so an operator who later retrieves them sees immediately that they were already investigated.

Separating a tag's meaning from the rationale for each application lets a later operator understand both what a group represents and why each member belongs to it, without repeating the analysis. The recorded author and timestamp indicate whether a finding is recent. Figs. 6–8 thus show how tagging supports collaboration among operators, fulfilling R3.

V. Evaluation

We evaluate SSHVisual's ability to uncover patterns and show hidden entities in two ways: 1) Security analysts finding campaigns in which the same usernames recur across IPs in different countries (Section V-A) and 2) Network administrators finding entities to block that act alike but are buried underneath others (Section V-B). Both subsections start with an operator finding an IP, e.g., from an intrusion detection alert, an abuse report, or the discovery tables in SSHVisual. On its own, such an IP is one address among many; SSHVisual helps the operator find what else it is connected to.

A. Username Campaigns

Fig. 9 shows the username search (label A) for IP 68.183.213.225 (DigitalOcean, Germany). Five IPs have 100% username similarity with it (labels B–F): 143.244.183.240, 24.199.108.32, and 146.190.113.228 in the US, 134.122.62.125 in the Netherlands, and 101.204.27.20 in China. All six try the same 77 usernames, and every list contains all 16 of the searched IP's uncommon usernames, such as ecell, guojing, and xxaifwq. These uncommon usernames make the match meaningful. A 100% score built on common usernames like root and admin would say little, but 16 shared uncommon usernames are unlikely by chance. The six IPs share two ASNs but span four countries, consistent with a single operator renting servers in several regions. The campaign also extends further: at least 12 IPs score above 84% similarity, spanning five countries and six ASNs (Appendix B).

Fig. 10 shows why this campaign may not be seen from volume. Each IP is active on different days: 146.190.113.228 on Dec 19 (label D), 143.244.183.240 on Dec 20 (label B), 24.199.108.32 on Dec 28 (label C), 68.183.213.225 on Jan 2 and Jan 8 (label A), 134.122.62.125 on Jan 5 to 6 (label E), and 101.204.27.20 on Nov 25 (label F). The lines never rise together, so the chart suggests six unrelated bursts. Shared usernames can link them into one campaign.

B. Finding Hidden Entities to Block

Fig. 11 shows the IP similarity search for 200.195.169.59, an IP from Ligga Telecomunicaç oes S.A. in Brazil with 36 thousand attacks. Its five closest matches (88.6% to 94.9% similarity) come from Brazil, Indonesia, Italy, Mexico, and Australia, each from a different ASN. Despite this spread, they behave alike. Each made roughly 30-40 thousand attacks and is tagged as exploratory targeting, meaning they try many different usernames. Each shares 1,684 to 1,899 usernames with the searched IP, 46 to 140 of them uncommon (tried by fewer than 1% of all IPs). The searched IP itself tried 3,821 usernames, 925 of them uncommon. The pattern extends beyond these five: the top 20 matches, all at least 86% similar, span 14 countries and 20 ASNs (Appendix C). Together they

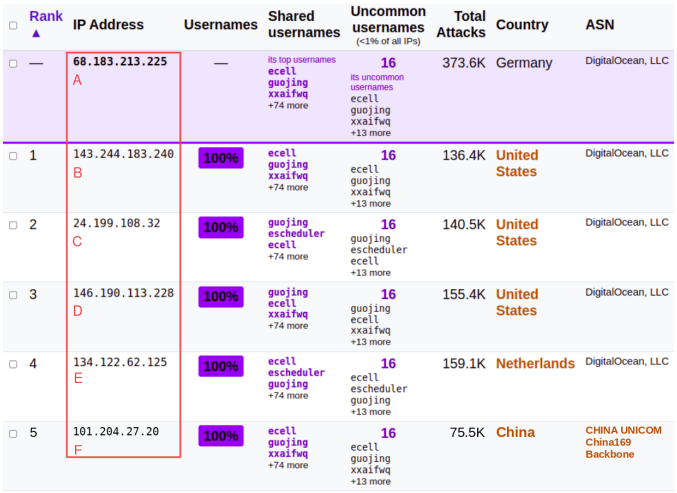


Fig. 9. 100% usernames shared among six IPs. IPs A, B, C, D, E, and F match the IPs A, B, C, D, E, and F in Fig. 10.

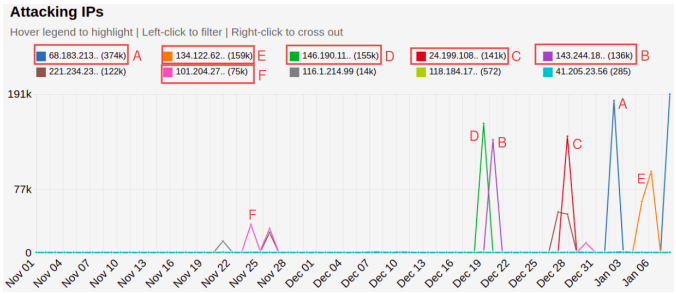


Fig. 10. Six IPs labeled A–F from Fig. 9 in the linechart, along with other IPs with almost the same username list.

show low-volume IPs in many countries working through large, overlapping lists of usernames at the same time.

Fig. 12 shows how buried these IPs are despite their interesting nature to administrators. None of the 10 IPs (selected from Fig. 11) would appear in the top volume IPs chart. The searched IP ranks #296 of 86,985 IPs and accounts for 0.018% of 206.6M attacks. Its country, Brazil, is large enough to appear in the top 10 countries (#8), but its ASN is 13 times too small to appear in the top 10 ASNs, and the IP itself is 20 times too small to appear in the top 10 IPs. The other IPs rank between #222 and #2,826. An administrator looking only at the high volume charts would see Brazil but never these IPs. With SSHVisual, administrators know not just the IP they found to block but also similar IPs that they should block, and most importantly, why they should block them.

VI. Conclusion

We presented SSHVisual, a visual analytics system that treats the username as a first-class analytic dimension alongside the IP, and ASN in SSH attack data. SSHVisual pairs a discovery table that ranks every entity with coordinated linecharts that follow Shneiderman's mantra, so operators can move between precise lookup and temporal behavior. Similarity search groups attackers by behavior along the

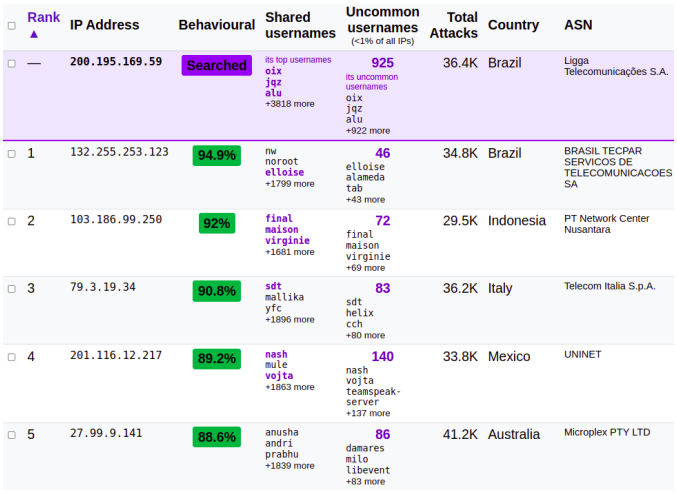


Fig. 11. Five IPs that behave like 200.195.169.59.

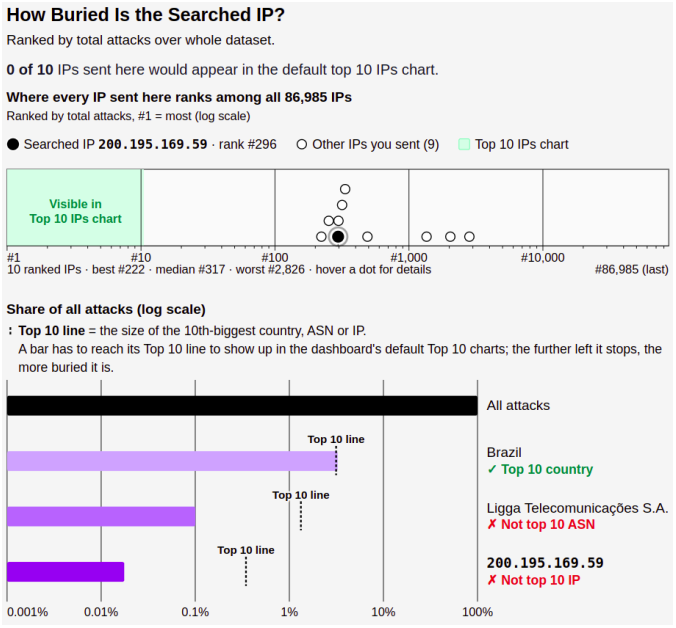


Fig. 12. How buried 200.195.169.59 and its similar IPs are.

username and IP dimensions and explains each match. Shared tagging preserves those groups so that findings persist and remain visible across operators. Our case studies show that this combination surfaces patterns that volume-based, IP-centric views can miss. Low-volume IPs can spread across many countries and ASNs worked through similar lists of uncommon usernames, yet none ranked high enough to appear in any top-ten chart. IPs that never attacked on the same day were linked into a single campaign only by the 77 usernames they shared. Because SSH traffic may not offer a normal baseline for automated detection, these results suggest that exploratory, operator-driven analysis remains a necessary complement to automated methods, and that the username is a valuable lens for recognizing coordinated attackers.

REFERENCES

[1] T. Ylonen, "Ssh–secure login connections over the internet," in *Proceedings of the 6th USENIX Security Symposium*, vol. 37, 1996, pp. 40–52.

[2] S. K. Singh, S. Gautam, C. Cartier, S. Patil, and R. Ricci, "Where the wild things are: Brute-Force SSH attacks in the wild and how to stop them," in *21st USENIX Symposium on Networked Systems Design and Implementation (NSDI 24)*. Santa Clara, CA: USENIX Association, Apr. 2024, pp. 1731–1750. [Online]. Available: https://www.usenix.org/conference/nsdi24/presentation/singh-sachin

[3] M. Antonakakis, T. April, M. Bailey, M. Bernhard, E. Bursztein, J. Cochran, Z. Durumeric, J. A. Halderman, L. Invernizzi, M. Kallitsis *et al.*, "Understanding the mirai botnet," in *26th USENIX security symposium (USENIX Security 17)*, 2017, pp. 1093–1110.

[4] C. Kolias, G. Kambourakis, A. Stavrou, and J. Voas, "Ddos in the iot: Mirai and other botnets," *computer*, vol. 50, no. 7, pp. 80–84, 2017.

[5] F. Dang, Z. Li, Y. Liu, E. Zhai, Q. A. Chen, T. Xu, Y. Chen, and J. Yang, "Understanding fileless attacks on linux-based iot devices with honeycloud," in *Proceedings of the 17th Annual International Conference on Mobile Systems, Applications, and Services*, 2019, pp. 482–493.

[6] S. Pastrana and G. Suarez-Tangil, "A first look at the crypto-mining malware ecosystem: A decade of unrestricted wealth," in *Proceedings of the Internet Measurement Conference*, 2019, pp. 73–86.

[7] V. T. Guimarães, C. M. D. S. Freitas, R. Sadre, L. M. R. Tarouco, and L. Z. Granville, "A survey on information visualization for network and service management," *IEEE Communications Surveys & Tutorials*, vol. 18, no. 1, pp. 285–323, 2016.

[8] H. Shiravi, A. Shiravi, and A. A. Ghorbani, "A survey of visualization systems for network security," *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 8, pp. 1313–1329, 2012.

[9] T. Zhang, X. Wang, Z. Li, F. Guo, Y. Ma, and W. Chen, "A survey of network anomaly visualization," *Science China Information Sciences*, vol. 60, no. 12, p. 121101, 2017.

[10] J. Raynor, T. Crnovrsanin, S. Di Bartolomeo, L. South, D. Saffo, and C. Dunne, "The state of the art in bgp visualization tools: A mapping of visualization techniques to cyberattack types," *IEEE Transactions on Visualization and Computer Graphics*, vol. 29, no. 1, pp. 1059–1069, 2022.

[11] C. Wu, S. Sheng, and X. Dong, "Research on visualization systems for ddos attack detection," in *2018 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. IEEE, 2018, pp. 2986–2991.

[12] C. Gates and C. Taylor, "Challenging the anomaly detection paradigm: A provocative discussion," in *Proceedings of the 2006 workshop on New security paradigms*, 2006, pp. 21–29.

[13] F. Mansmann and S. Mansmann, "Interactive exploration of data traffic with hierarchical network maps," 2006.

[14] J. Oberheide, M. Goff, and M. Karir, "Flamingo: Visualizing internet traffic," in *2006 IEEE/IFIP Network Operations and Management Symposium NOMS 2006*. IEEE, 2006, pp. 150–161.

[15] F. Mansmann, F. Fischer, D. A. Keim, and S. C. North, "Visual support for analyzing network traffic and intrusion detection events using treemap and graph representations," in *Proceedings of the Symposium on Computer Human Interaction for the Management of Information Technology*, 2009, pp. 19–28.

[16] T. Taylor, D. Paterson, J. Glanfield, C. Gates, S. Brooks, and J. McHugh, "Flovis: Flow visualization system," in *2009 Cybersecurity Applications & Technology Conference for Homeland Security*. IEEE, 2009, pp. 186–198.

[17] S. Chen, C. Guo, X. Yuan, F. Merkle, H. Schaefer, and T. Ertl, "Oceans: Online collaborative explorative analysis on network security," in *Proceedings of the Eleventh Workshop on Visualization for Cyber Security*, 2014, pp. 1–8.

[18] T. Tremel, J. Kogel, F. Jauernig, S. Meier, D. Thom, F. Becker, C. Muller, and S. Koch, "Vitalflow: Visual interactive traffic analysis with netflow," in *NOMS 2022-2022 IEEE/IFIP Network Operations and Management Symposium*. IEEE, 2022, pp. 1–6.

[19] L. Xiao, J. Gerth, and P. Hanrahan, "Enhancing visual analysis of network traffic using a knowledge representation," in *2006 IEEE symposium on visual analytics science and technology*. IEEE, 2006, pp. 107–114.

[20] J. R. Goodall and M. Sowul, "Viassist: Visual analytics for cyber defense," in *2009 IEEE conference on technologies for homeland security*. IEEE, 2009, pp. 143–150.

[21] D. H. Jeong, J.-H. Cho, F. Chen, L. Kaplan, A. Jøsang, and S.-Y. Ji, "Interactive web-based visual analysis on network traffic data," *Information*, vol. 14, no. 1, p. 16, 2022.

[22] H. Choi, H. Lee, and H. Kim, "Fast detection and visualization of network attacks on parallel coordinates," *computers & security*, vol. 28, no. 5, pp. 276–288, 2009.

[23] Y. Livnat, J. Agutter, S. Moon, and S. Foresti, "Visual correlation for situational awareness," in *IEEE Symposium on Information Visualization, 2005. INFOVIS 2005.* IEEE, 2005, pp. 95–102.

[24] C. Humphries, N. Prigent, C. Bidan, and F. Majorczyk, "Elvis: Extensible log visualization," in *Proceedings of the Tenth Workshop on Visualization for Cyber Security*, 2013, pp. 9–16.

[25] T. Yu, R. Lippmann, J. Riordan, and S. Boyer, "Ember: a global perspective on extreme malicious behavior," in *Proceedings of the seventh international symposium on visualization for cyber security*, 2010, pp. 1–12.

[26] D. H. Jeong, B.-K. Jeong, and S.-Y. Ji, "Multi-resolution analysis with visualization to determine network attack patterns," *Applied Sciences*, vol. 13, no. 6, p. 3792, 2023.

[27] A. King, B. Huffaker, A. Dainotti, and K. Claffy, "A coordinated view of the temporal evolution of large-scale internet events," *Computing*, vol. 96, no. 1, pp. 53–65, 2014.

[28] D. Ceneda, M. Di Bartolomeo, V. Di Donato, M. Patrignani, M. Pizzonia, and M. Rimondini, "Routingwatch: Visual exploration and analysis of routing events," in *NOMS 2016-2016 IEEE/IFIP Network Operations and Management Symposium*. IEEE, 2016, pp. 591–597.

[29] J. C. Roberts, "State of the art: Coordinated & multiple views in exploratory visualization," in *Fifth international conference on coordinated and multiple views in exploratory visualization (CMV 2007)*. IEEE, 2007, pp. 61–71.

[30] J. S. Yi, Y.-a. Kang, J. T. Stasko, and J. A. Jacko, "Understanding and characterizing insights: how do people gain insights using information visualization?" in *Proceedings of the 2008 Workshop on BEyond time and errors: novel evaLuation methods for Information Visualization*, 2008, pp. 1–6.

[31] R. Djap, C. Lim, K. E. Silaen, and A. Yusuf, "Xb-pot: Revealing honeypot-based attacker's behaviors," in *2021 9th International Conference on Information and Communication Technology (ICoICT)*. IEEE, 2021, pp. 550–555.

[32] V. Visoottiviseth, U. Jaralrungroj, E. Phoomrungraungsuk, and P. Kultanon, "Distributed honeypot log management and visualization of attacker geographical distribution," in *2011 Eighth International Joint Conference on Computer Science and Software Engineering (JCSSE)*. IEEE, 2011, pp. 23–28.

[33] M. Baykara and R. Das, "A novel honeypot based security approach for real-time intrusion detection and prevention systems," *Journal of Information Security and Applications*, vol. 41, pp. 103–116, 2018.

[34] A. Belqruch and A. Maach, "Scada security using ssh honeypot," in *Proceedings of the 2nd International Conference on Networking, Information Systems & Security*, 2019, pp. 1–5.

[35] I. Koniaris, G. Papadimitriou, and P. Nicopolitidis, "Analysis and visualization of ssh attacks using honeypots," in *Eurocon 2013*. IEEE, 2013, pp. 65–72.

[36] A. Kyriakou and N. Sklavos, "Container-based honeypot deployment for the analysis of malicious activity," in *2018 Global Information Infrastructure and Networking Symposium (GIIS)*. IEEE, 2018, pp. 1–4.

[37] I. Koniaris, G. Papadimitriou, P. Nicopolitidis, and M. Obaidat, "Honeypots deployment for the analysis and visualization of malware activity and malicious connections," in *2014 IEEE international conference on communications (ICC)*. IEEE, 2014, pp. 1819–1824.

[38] D. Fraunholz, M. Zimmermann, S. D. Anton, J. Schneider, and H. D. Schotten, "Distributed and highly-scalable wan network attack sensing and sophisticated analysing framework based on honeypot technology," in *2017 7th International Conference on Cloud Computing, Data Science & Engineering-Confluence*. IEEE, 2017, pp. 416–421.

[39] J. R. Goodall, E. D. Ragan, C. A. Steed, J. W. Reed, G. D. Richardson, K. M. Huffer, R. A. Bridges, and J. A. Laska, "Situ: Identifying and explaining suspicious behavior in networks," *IEEE transactions on visualization and computer graphics*, vol. 25, no. 1, pp. 204–214, 2018.

[40] Q. Liao and A. Striegel, "Intelligent network management using graph differential anomaly visualization," in *2012 IEEE Network Operations and Management Symposium*. IEEE, 2012, pp. 1008–1014.

[41] W. A. Pike, C. Scherrer, and S. Zabriskie, "Putting security in context: Visual correlation of network activity with real-world information," in *VizSEC 2007: Proceedings of the Workshop on Visualization for Computer Security*. Springer, 2008, pp. 203–220.
[42] L. Hao, C. G. Healey, and S. E. Hutchinson, "Ensemble visualization for cyber situation awareness of network security data," in *2015 IEEE Symposium on Visualization for Cyber Security (VizSec)*. IEEE, 2015, pp. 1–8.
[43] F. Pouget, G. Urvoy-Keller, and M. Dacier, "Time signatures to detect multi-headed stealthy attack tools," in *18th Annual FIRST Conference*, Baltimore, MD, USA, 2006.
[44] O. Thonnard and M. Dacier, "A framework for attack patterns' discovery in honeynet data," *Digital Investigation*, vol. 5, pp. S128–S139, 2008, proc. DFRWS 2008.
[45] R. Sommer and V. Paxson, "Outside the closed world: On using machine learning for network intrusion detection," in *2010 IEEE symposium on security and privacy*. IEEE, 2010, pp. 305–316.
[46] S. Axelsson, "The base-rate fallacy and the difficulty of intrusion detection," *ACM Transactions on Information and System Security (TISSEC)*, vol. 3, no. 3, pp. 186–205, 2000.
[47] D. M. Best, R. P. Hafen, B. K. Olsen, and W. A. Pike, "Atypical behavior identification in large-scale network traffic," in *2011 IEEE Symposium on Large Data Analysis and Visualization*. IEEE, 2011, pp. 15–22.
[48] Y. Xie, F. Yu, K. Achan, E. Gillum, M. Goldszmidt, and T. Wobber, "How dynamic are ip addresses?" in *Proceedings of the 2007 conference on Applications, technologies, architectures, and protocols for computer communications*, 2007, pp. 301–312.
[49] B. Stone-Gross, M. Cova, L. Cavallaro, B. Gilbert, M. Szydlowski, R. Kemmerer, C. Kruegel, and G. Vigna, "Your botnet is my botnet: analysis of a botnet takeover," in *Proceedings of the 16th ACM conference on Computer and communications security*, 2009, pp. 635–647.
[50] G. C. Moura, C. Ganán, Q. Lone, P. Poursaied, H. Asghari, and M. van Eeten, "How dynamic is the isps address space? towards internet-wide dhcp churn estimation," in *2015 IFIP Networking Conference (IFIP Networking)*. IEEE, 2015, pp. 1–9.
[51] J. L. Guerra, E. Veas, and C. A. Catania, "A study on labeling network hostile behavior with intelligent interactive tools," in *2019 IEEE Symposium on Visualization for Cyber Security (VizSec)*. IEEE, 2019, pp. 1–10.
[52] J. Owens and J. Matthews, "A study of passwords and methods used in brute-force SSH attacks," in *USENIX Workshop on Large-Scale Exploits and Emergent Threats (LEET)*, San Francisco, CA, USA, 2008. [Online]. Available: https://lin-web.clarkson.edu/~jmatthew/publications/leet08.pdf
[53] Y. Wu, P. M. Cao, A. Withers, Z. T. Kalbarczyk, and R. K. Iyer, "Mining threat intelligence from billion-scale SSH brute-force attacks," in *Workshop on Decentralized IoT Systems and Security (DISS)*. San Diego, CA, USA: Internet Society, 2020. [Online]. Available: https://www.ndss-symposium.org/wp-content/uploads/2020/04/diss2020-23007-paper.pdf
[54] B. Shneiderman, "The eyes have it: A task by data type taxonomy for information visualizations," in *Proceedings of the IEEE Symposium on Visual Languages*, 1996, pp. 336–343.
[55] W. Javed, B. McDonnel, and N. Elmqvist, "Graphical perception of multiple time series," *IEEE Transactions on Visualization and Computer Graphics*, vol. 16, no. 6, pp. 927–934, 2010.

APPENDIX A
DISCOVERY TABLE COLUMNS

This appendix lists every column of the discovery table described in Section IV-A1. That section shows the Countries tab (Fig. 1) and describes only the columns shared by all four tabs (label C). The remaining columns are selected from the drop-down menu (label D) and depend on the tab. Table I lists every column and the tabs in which it appears, and Figs. 13–16 show all columns of the Countries, IP Addresses, ASN Organizations, and Usernames tabs, respectively.

Columns shared by all tabs. The upper part of Table I lists the ten columns described in Section IV-A1: total attacks, average daily attacks, persistence, the largest absolute and percentage daily change, attacks in the most recent seven days, first- and last-seen dates, maximum daily count, trend, and burst intensity.

Cross-dimension columns. The lower part of Table I lists columns that describe an entity in terms of another dimension. For each of the four dimensions (usernames, IPs, ASNs, and countries), a tab can have three such columns:

- *Unique count*: the number of distinct values of that dimension associated with the entity, e.g., how many distinct usernames a country's IPs tried.
- *Stability*: the mean Jaccard similarity between consecutive days' sets of that dimension, from 0 (volatile) to 1 (stable). A higher value means the entity uses the same values from day to day.
- *Concentration*: the top three values of that dimension and their percentages of the entity's attacks.

A tab never has these columns for its own dimension; for example, the Usernames tab has no username stability column. The IP tab also has no ASN or country columns of this kind, because each IP address is assumed to belong to exactly one ASN and one country.

TABLE I
DISCOVERY TABLE COLUMNS BY TAB

Column	Countries	IPs	ASNs	Usernames
Total attacks	✓	✓	✓	✓
Average daily	✓	✓	✓	✓
Persistence	✓	✓	✓	✓
Max absolute Δ	✓	✓	✓	✓
Max % Δ	✓	✓	✓	✓
Recent (7d)	✓	✓	✓	✓
First / last seen	✓	✓	✓	✓
Max daily	✓	✓	✓	✓
Trend	✓	✓	✓	✓
Burst intensity	✓	✓	✓	✓
Unique usernames	✓	✓	✓	
Unique IPs	✓		✓	✓
Unique ASNs	✓			✓
Unique Countries				✓
Username stability	✓	✓	✓	
IP stability	✓		✓	✓
ASN stability	✓			✓
Country stability			✓	✓
Username concentration	✓	✓	✓	
IP concentration	✓		✓	✓
ASN concentration	✓			✓
Country concentration			✓	✓

APPENDIX B
USERNAME CAMPAIGNS

This appendix extends the username campaign in Section V-A. There, the username similarity search for 68.183.213.225 (DigitalOcean, Germany) returned five IPs with 100% username similarity (Fig. 9). All six IPs tried the same 77 usernames, including 16 uncommon ones. Fig. 17 shows the same search beyond those six results. 12 IPs score above 84% similarity, and they span five countries and six ASNs. The 100% matches (except one) all share one ASN, but the wider set reaches other providers. This suggests that

Rank	Country	Total Attacks	Avg Daily	Persistence	Max Absolute Δ	Max % Δ	Recent (7d)	First Seen	Last Seen	Max Daily	Unique ASNs	Unique IPs	Unique Usernames	ASN Stability	IP Stability	Username Stability	ASN Concentration (Top 3)	IP Concentration (Top 3)	Username Concentration (Top 3)	Trend (7-day)	Burst Intensity
1	United States Spray	38.2M	553.0K	100.00% (69d)	748.4K	342.86%	6.0M	Nov 1, 2022	Jan 8, 2023	1.6M	661	13.2K	82.0K	0.705	0.711	0.479	DigitalOcean, LLC (42.9%) Tencent Building, Kejizhongyi Avenue (7.3%) Microsoft Corporation (5.3%)	209.239.112.213 (3.3%) 23.224.95.63 (1.9%) 98.159.98.150 (1.6%)	root (45.6%) admin (3.1%) ubuntu (1.7%)		3.0x
2	China Spray	28.3M	410.1K	100.00% (69d)	1.1M	270.43%	2.0M	Nov 1, 2022	Jan 8, 2023	1.5M	117	10.2K	34.7K	0.814	0.498	0.418	CHINANET-BACKBONE (48.8%) Shenzhen Tencent Computer Systems Company Limited (18.1%) CHINA UNICOM China169 Backbone (7.9%)	180.96.11.56 (2.8%) 61.177.173.53 (2.5%) 61.177.173.49 (2.4%)	root (73.6%) admin (2.9%) user (2.2%)		3.6x
3	Singapore Spray	13.9M	201.2K	100.00% (69d)	850.7K	3686.05%	2.4M	Nov 1, 2022	Jan 8, 2023	873.7K	72	3.7K	51.4K	0.848	0.763	0.466	DigitalOcean, LLC (59.2%) Tencent Building, Kejizhongyi Avenue (22.3%) GoDaddy.com, LLC (3.5%)	43.156.30.199 (6.2%) 148.72.247.138 (1.4%) 157.230.244.66 (0.5%)	root (48.9%) admin (2.1%) ubuntu (1.5%)		4.3x

Fig. 13. All the columns of the Countries tab

Rank	IP Address	Total Attacks	Avg Daily [1]	Persistence [2]	Max Absolute Δ	Max % Δ	Recent (7d)	First Seen	Last Seen	Max Daily	Unique Usernames	Username Concentration (Top 3)	Username Stability	Trend (7-day)	Burst Intensity
1	61.177.173.53	695.5K	10.1K	100.00% (69d)	8.1K	152.20%	36.9K	Nov 1, 2022	Jan 8, 2023	18.2K	1	root (100.0%)	1.000		1.8x
2	61.177.173.49	674.0K	9.8K	100.00% (69d)	9.1K	15.44K%	40.4K	Nov 1, 2022	Jan 8, 2023	20.7K	1	root (100.0%)	1.000		2.1x
3	61.177.173.47	633.0K	9.2K	100.00% (69d)	7.4K	111.83%	36.7K	Nov 1, 2022	Jan 8, 2023	16.6K	1	root (100.0%)	1.000		1.8x
4	61.177.173.50	611.8K	8.9K	100.00% (69d)	10.5K	26.95K%	34.8K	Nov 1, 2022	Jan 8, 2023	20.4K	1	root (100.0%)	1.000		2.3x

Fig. 14. All the columns of the IP Addresses tab

Rank	ASN	Total Attacks	Avg Daily	Persistence	Max Absolute Δ	Max % Δ	Recent (7d)	First Seen	Last Seen	Max Daily	Country Concentration (Top 3)	Country Stability	Unique IPs	IP Concentration (Top 3)	Unique Usernames	Username Concentration (Top 3)	IP Stability	Username Stability	Trend (7-day)	Burst Intensity
1	DigitalOcean, LLC	42.4M	80.8K	100.00% (69d)	810.8K	251.22K%	8.1M	Nov 1, 2022	Jan 8, 2023	829.6K	United States (38.6%) Singapore (19.4%) India (13.0%)	0.987	6.0K	68.183.213.225 (0.9%) 143.244.158.100 (0.6%) 68.183.221.15 (0.4%)	61.4K	root (45.3%) admin (1.9%) ubuntu (1.9%)	0.862	0.537		10.3x
2	CHINANET-BACKBONE	13.8M	200.2K	100.00% (69d)	553.1K	329.67%	775.5K	Nov 1, 2022	Jan 8, 2023	751.5K	China (100.0%)	1.000	3.5K	61.177.173.53 (5.0%) 61.177.173.49 (4.9%) 61.177.173.46 (4.8%)	1.7K	root (91.2%) admin (1.2%) user (0.8%)	0.416	0.578		3.8x
3	Tencent Building, Kejizhongyi Avenue	8.6M	13.2K	100.00% (69d)	853.5K	389.72K%	1.5M	Nov 1, 2022	Jan 8, 2023	853.7K	Singapore (35.9%) United States (32.4%) Germany (14.0%)	0.967	1.7K	43.156.30.199 (10.0%) 49.51.19.119 (0.5%) 43.131.41.251 (0.4%)	47.8K	root (51.3%) admin (1.5%) ubuntu (1.3%)	0.805	0.422		64.6x

Fig. 15. All the columns of the ASN Organizations tab

the campaign's username list is used beyond a single hosting provider, either by one operator renting servers from several providers or by several operators sharing a list.

APPENDIX C
HIDDEN ENTITIES

This appendix extends the IP similarity search in Section V-B. There, the five closest matches to 200.195.169.59 (Ligga Telecomunicações S.A., Brazil) were each at least 88.6% similar and each came from a different country and ASN (Fig. 11). Fig. 18 shows the top 20 matches. All are at least 86% similar, and they span 14 countries and 20 ASNs, so no two matches share an ASN. Blocking by country or by ASN would therefore not catch this group. The similarity search gives administrators the full set of IPs to block, along with the shared behavior that justifies blocking them.

Rank	Username	Total Attacks	Avg Daily	Persistence	Max Absolute Δ	Max % Δ	Recent (7d)	First Seen	Last Seen	Max Daily	Unique Countries	Unique ASNs	Unique IPs	Country Concentration (Top 3)	ASN Concentration (Top 3)	IP Concentration (Top 3)	Country Stability	ASN Stability	IP Stability	Trend (7-day)	Burst Intensity
1	root	97.5M	1.4M	100.00% (69d)	1.8M	194.38%	10.8M	Nov 1, 2022	Jan 8, 2023	5.2M	192	4.5K	67.5K	China (21.4%) United States (17.8%) Singapore (7.0%)	DigitalOcean, LLC (19.7%) CHINANET-BACKBONE (12.9%) Tencent Building, Kejizhongyi Avenue (4.5%)	86.51.167.31 (1.0%) 43.156.30.199 (0.9%) 180.96.11.56 (0.8%)	0.906	0.692	0.590		3.7x
2	admin	6.9M	99.4K	100.00% (69d)	83.6K	140.68%	919.0K	Nov 1, 2022	Jan 8, 2023	195.5K	191	4.1K	47.7K	United States (17.3%) Russia (13.2%) China (12.0%)	DigitalOcean, LLC (12.0%) NONE (6.4%) Google LLC (4.9%)	91.212.166.22 (3.0%) 152.89.196.220 (2.6%) 152.89.196.123 (2.6%)	0.831	0.541	0.371		2.0x
3	user	4.5M	64.6K	100.00% (69d)	61.2K	151.86%	499.5K	Nov 1, 2022	Jan 8, 2023	142.1K	188	3.9K	41.7K	Lithuania (15.2%) China (14.0%) United States (13.8%)	UAB Host Baltic (15.2%) DigitalOcean, LLC (13.1%) OVH SAS (3.8%)	141.98.11.30 (9.7%) 141.98.10.210 (3.5%) 179.60.147.101 (1.5%)	0.805	0.499	0.332		2.2x

Fig. 16. All the columns of the Usernames tab

Rank	IP Address	Usernames	Shared usernames	Uncommon usernames (<1% of all IPs)	Total Attacks	Country	ASN
—	68.183.213.225	—	77	16	373.6K	Germany	DigitalOcean, LLC
1	143.244.183.240	100%	77	16	136.4K	United States	DigitalOcean, LLC
2	24.199.108.32	100%	77	16	140.5K	United States	DigitalOcean, LLC
3	146.190.113.228	100%	77	16	155.4K	United States	DigitalOcean, LLC
4	134.122.62.125	100%	77	16	159.1K	Netherlands	DigitalOcean, LLC
5	101.204.27.20	100%	77	16	75.5K	China	CHINA UNICOM China169 Backbone
6	116.1.214.99	99.9%	77	16	13.9K	China	CHINANET-BACKBONE
7	221.234.230.12	99.8%	77	16	122.4K	China	CHINANET-BACKBONE
8	118.184.171.58	98.1%	77	16	572	China	ChangZhou Bitcomm Software Technology Co., Limited
9	41.205.23.56	94.7%	72	16	285	Cameroon	MTN Network Solutions (Cameroon)
10	211.155.104.65	93.4%	77	16	1.4K	China	Kaopu Cloud HK Limited
11	104.248.205.76	85.8%	54	11	121.6K	Netherlands	DigitalOcean, LLC
12	68.183.221.15	84.9%	77	16	181.3K	Germany	DigitalOcean, LLC

Fig. 17. Username similarity search for 68.183.213.225 (Section V-A), showing IPs above 84% similarity across five countries and six ASNs.

Rank	IP Address	Behavioural	Shared usernames	Uncommon usernames (<1% of all IPs)	Total Attacks	Country	ASN
—	200.195.169.59	Searched	3,821	925	36.4K	Brazil	Ligga Telecomunicações S.A.
1	132.255.253.123	94.9%	1,802	46	34.8K	Brazil	BRASIL TECPAR SERVICOS DE TELECOMUNICACOES SA
2	103.186.99.250	92%	1,684	72	29.5K	Indonesia	PT Network Center Nusantara
3	79.3.19.34	90.8%	1,899	83	36.2K	Italy	Telecom Italia S.p.A.
4	201.116.12.217	89.2%	1,866	140	33.8K	Mexico	UNINET
5	27.99.9.141	88.6%	1,842	86	41.2K	Australia	Microplex PTY LTD
6	45.118.145.221	87.5%	1,525	52	26.8K	Vietnam	Long Van Soft Solution JSC
7	103.110.8.244	87.4%	1,480	57	24.9K	Indonesia	PT Citra Jelajah Informatika
8	160.120.247.113	86.9%	1,480	40	22.5K	Ivory Coast	Orange Côte d'Ivoire
9	211.46.4.196	86.9%	1,907	77	44.2K	South Korea	Korea Telecom
10	77.235.29.116	86.9%	1,526	58	25.8K	Kyrgyzstan	Mega-Line Ltd.
11	130.61.35.0	86.8%	1,554	58	27.0K	Germany	Oracle Corporation
12	202.4.42.154	86.7%	1,756	144	32.6K	Samoa	Vodafone Samoa Limited
13	43.131.43.132	86.7%	1,609	58	27.0K	Germany	Tencent Building, Kejizhongyi Avenue
14	82.160.101.186	86.7%	1,769	99	27.7K	Poland	TK Telekom sp. z o.o.
15	188.250.172.49	86.4%	1,654	49	29.8K	Portugal	MEO - SERVICOS DE COMUNICACOES E MULTIMEDIA S.A.
16	200.53.21.177	86.4%	1,889	78	39.6K	Brazil	ACESSOLINE TELECOM ACESSO (ALT)
17	202.57.17.34	86.4%	1,877	99	39.2K	Indonesia	PT. Sejuta Jaring Global
18	2.40.59.98	86.3%	1,648	47	27.9K	Italy	Vodafone Italia S.p.A.
19	81.219.237.226	86.1%	1,717	55	32.3K	Poland	Netia SA
20	188.68.31.75	86%	1,587	51	26.5K	Russia	LTD AtelRybinsk

Fig. 18. Top 20 IP similarity matches for 200.195.169.59 (Section V-B), spanning 14 countries and 20 ASNs.